Quantum Geometric Tensor of the SU(1,1) and SU(2) groups

Lévay Péter, Emmer Marcell
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I. QGT SUMMARY [?]

We represent a quantum state by the wavefunction

$$|\psi\rangle = \sum_k C_n |e_n\rangle \tag{1}$$

where $C_n = [C_1, C_2, ..., C_N]$ is a complex vector and $|e_n\rangle$ is an ONB. In this case the QGT is nothing else but the Fubini-Study metric (FSM) on the CP^N manifold.

Considering a family of parametrised Hamiltonians $\{H(X)\}$ with eigenenergies $E_n(X)$ and eigenstates $|\phi_n(X)\rangle$. The system is in state $|\psi(X)\rangle$. With it we define a quantum distance by varying the parameter dX.

$$\begin{aligned} ds^2 &= ||\psi(X+dX) - \psi(X)||^2 \\ &= \langle \partial_j \psi | \partial_k \psi \rangle \, dX^j dX^k = \gamma_{jk}(X) dX^j dX^k \end{aligned} \tag{2}$$

Where $\gamma_{jk}(X)$ is a metric but not gauge-invariant. The gauge-invariant metric is the following

$$g_{jk}(X) = \gamma_{jk}(X) - \beta_j(X)\beta_k(X) \tag{3a}$$
$$\beta_j(X) = i \langle \psi(X) | \partial_j \psi(X) \rangle \tag{3b}$$

It measures the distance of "Rays" in the projected Hilbert space $\mathcal{PH} = \mathcal{H}/U(1)$. We define the QGT, which is the Fubini-Study metric on quantum Rays, as

$$\begin{aligned} T_{jk}(X) = &\langle \partial_j \psi(X) | \partial_k \psi(X) \rangle \\ &- \langle \partial_j \psi(X) | \psi(X) \rangle \langle \psi(X) | \partial_k \psi(X) \rangle \end{aligned} \tag{4}$$

In other ways we think of our state as initial state $|n\rangle_0$ acted upon by an $O(X)$ operator. $|n\rangle_0$ is the eigenstate of the Hamiltonian living in the origin of the parameter space $X = 0$ while the $O(X)$ operator contains the information about where the state is in this parameter space $X \in \mathcal{M}$ and also serves as a representation of the system's symmetry group. In this case the QGT is the following.

$$T^{(n)}_{jk}(X) = \sum_{m \neq n} \langle \partial_j n | m \rangle \langle m | \partial_k n \rangle \tag{5}$$
$$|n\rangle = |n(X)\rangle = O(X) |n\rangle_0 \tag{6}$$

II. QGT FOR THE TWO-OSCILLATOR REALIZATION OF SU(1,1)

The $\mathfrak{su}(1,1)$ Lie-algebra has 3 generators K_i $(i = 1,2,3)$ obeying the following commutation identities

$$[K_i, K_j] = -i\epsilon_{ijk} K^k \tag{7}$$

With the metric of $G_{\mu\nu} = G^{\mu\nu} = \mathrm{diag}(-1,-1,1)$.

In the Schwinger-type two-boson realization of $\mathfrak{su}(1,1)$ the generators are the following

$$K_1 = -\frac{1}{2}\left(a^\dagger b^\dagger + ab\right) \tag{8a}$$
$$K_2 = -\frac{i}{2}\left(ab - a^\dagger b^\dagger\right) \tag{8b}$$
$$K_3 = \frac{1}{2}\left(a^\dagger a + b^\dagger b + 1\right) \tag{8c}$$
$$\left[a, a^\dagger\right] = \left[b, b^\dagger\right] = 1 \tag{8d}$$

This is an infinite dimensional representation of $SU(1,1)$ and these generators are hermitian. We would like to construct the CGT of the $SU(1,1)$ CSs defined in the Perelomovian way [?]. In this case our initial state is the GS of the separable 2-oscillator Hamiltonian $|0,0\rangle_0$ and the unitary CS operator is

$$O(\theta,\varphi) = \exp\left[i\theta\left(-\sin\varphi K_1 + \cos\varphi K_2\right)\right] \tag{9a}$$
$$\eta = \tanh(\theta/2)\exp(i\varphi) \tag{9b}$$

To be more general, instead of $|0,0\rangle_0$ we take our initial state to be $|n_1, n_2\rangle_0$, the eigenstate of the uncoupled 2 oscillator Hamiltonian K_3, and the quantum numbers n_1, n_2 correspond to the two excitation numbers. In this way, we can construct $T^{(n_1,n_2)}_{jk}$ and not just $T^{(0,0)}_{jk}$.

First let's write the expression inside the sum of the CGT the following way

$$\langle m_1, m_2 | \partial_k n_1, n_2 \rangle =_0 \langle m_1, m_2 | O^{-1} \partial_k O | n_1, n_2 \rangle_0 \tag{10}$$

And we find out what are the coefficients of $O^{-1}\partial_k O$ in the basis of $\{K_i\}$.

$$O^{-1}\partial_k O = E^1_k K_1 + E^2_k K_2 + A^3_k K_3 \quad (k = \theta, \varphi) \tag{11}$$

The E^a_k ad A^3_k coefficients are independent of representation so we introduce a different rep of SU(1,1) made up of 2×2 matrices.

$$K_1 = \frac{i}{2}\sigma_1 \quad K_1 = \frac{i}{2}\sigma_2 \quad K_3 = \frac{1}{2}\sigma_3 \tag{12}$$

Since this is a finite dimensional representation of $SU(1,1)$ it could only be non-hermitian. Therefore,

$O(\theta,\varphi)$ wont be unitary anymore. In this basis, one can prove that it has the following form.

$$O(\theta,\varphi) = \begin{bmatrix} \cosh\frac{\theta}{2} & i\sinh\frac{\theta}{2}e^{-i\varphi} \\ -i\sinh\frac{\theta}{2}e^{i\varphi} & \cosh\frac{\theta}{2} \end{bmatrix} \tag{13}$$

From this it's easy to see that the coefficients are

$$E^1_\theta = -i\sin\varphi \qquad E^1_\varphi = -i\sinh\theta\cos\varphi \tag{14a}$$
$$E^2_\theta = i\cos\varphi \qquad E^2_\varphi = -i\sinh\theta\sin\varphi \tag{14b}$$
$$A^3_\theta = 0 \qquad A^3_\varphi = -2i\sinh^2\frac{\theta}{2} \tag{14c}$$

Since $|n_1,n_2\rangle$ is an eigenstate of K_3 in the Schwinger-type two-boson realization, and $m_i \neq n_i$ in the summation of the CGT's defining equation (5), the only relevant coefficients are of the index $a = 1,2$. The SU(1,1) CGT can be rewritten into the following form using the identity $\mathbb{1} = \sum_{m_1,m_2} |m_1,m_2\rangle_0 \langle m_1,m_2|$

$$T^{(n_1,n_2)}_{jk}(\theta,\varphi) = -\sum_{a,b=1,2} E^a_j(\theta,\varphi)E^b_k(\theta,\varphi) \times {}_0\langle n_1,n_2| K_aK_b |n_1,n_2\rangle_0 \tag{15}$$

Since $T^{(n_1,n_2)}_{jk}$ is hermitian, it can be written up with the real valued matrices $g^{(n_1,n_2)}_{jk}$ and $\Omega^{(n_1,n_2)}_{jk}$

$$T^{(n_1,n_2)}_{jk} = g^{(n_1,n_2)}_{jk} + i\Omega^{(n_1,n_2)}_{jk} \tag{16a}$$
$$g^{(n_1,n_2)T}_{jk} = g^{(n_1,n_2)}_{jk} \tag{16b}$$
$$\Omega^{(n_1,n_2)T}_{jk} = -\Omega^{(n_1,n_2)}_{jk} \tag{16c}$$

$\Omega^{(n_1,n_2)}_{jk}$ corresponds to the antisymmetric commutator part and $g^{(n_1,n_2)}_{jk}$ corresponds to the symmetric anticommutator part of the resolution in equation (15). Both the real and imaginary part of the CGT has a physical meaning. We can think of $g^{(n_1,n_2)}_{jk}$ as the metric tensor of the manifold generated by the action of the Lie group $SU(1,1)$ on a fixed quantum state [**?**]. Following Perelomov [6] we call them generalized coherent states manifolds (G.C.S.M.). The components of such metric tensor are related to the dispersion of the quantum operators acting on the underlying Hilbert space. In a similar way, $\Omega^{(n_1,n_2)}_{jk}$ corresponds to a complex metric's imaginary part.

$$K_aK_b = \frac{1}{2}[K_a,K_b] + \frac{1}{2}\{K_a,K_b\} \tag{17}$$

After some calculation, we can see that g and Ω have the following form.

$$g^{(n_1,n_2)}_{jk} = (\Delta K_1)^2_0 \begin{bmatrix} 1 & 0 \\ 0 & \sinh^2\theta \end{bmatrix} \tag{18a}$$
$$(\Delta K_1)^2_0 = (\Delta K_2)^2_0 = {}_0\langle n_1,n_2| K_1^2 |n_1,n_2\rangle_0 = \frac{1}{4}(n_1+1)(n_2+1) \tag{18b}$$

$$\Omega^{(n_1,n_2)}_{jk} = \langle K_3\rangle_0 \begin{bmatrix} 0 & -\frac{1}{2}\sinh\theta \\ \frac{1}{2}\sinh\theta & 0 \end{bmatrix} \tag{19a}$$
$$\langle K_3\rangle_0 = {}_0\langle n_1,n_2| K_3 |n_1,n_2\rangle_0 = \frac{1}{2}(n_1+n_2+1) \tag{19b}$$

To see in detail how the metric tensor is related to quantum fluctuations, we calculate the determinant of $T^{(n)}_{jk}$.

$$\det T^{(n_1,n_2)}_{jk} = \left[(\Delta K_1)^2_0(\Delta K_1)^2_0 - \left|\frac{1}{2}\langle K_3\rangle_0\right|^2\right]\sinh\theta \tag{20}$$

Where the Heisenberg uncertainty principle tells us that the expression inside the square bracket has to be greater than zero.

$$(\Delta K_1)_0(\Delta K_2)_0 \geq \frac{1}{2}|\langle K_3\rangle_0| \tag{21}$$

In the case of the 2 oscillator realization the determinant has the following form.

$$\det T^{(n_1,n_2)}_{jk} = \frac{1}{8}n_1n_2\left(\frac{1}{2}n_1n_2 + n_1 + n_2 + 1\right)\sinh\theta \tag{22}$$

Which equal zero if either of the two oscillators is in the ground state, meaning $n_1 = 0$ or $n_2 = 0$, or in the case of the $\theta = 0$ separable case. So the Perelomovian CSs, where $n_1 = n_2 = 0$ but $\theta \neq 0$, the QGT is non-invertible, therefore it is not a proper metric.

III. QGT FOR THE ONE-OSCILLATOR REALIZATION OF SU(1,1)

For a single-mode field described by the annihilation and creation operators a and a, respectively, the Lie $\mathfrak{su}(1,1)$ algebra is realized by the operators

$$K_3 = \frac{1}{2}(a^\dagger a + \frac{1}{2}) \qquad K_- = \frac{1}{2}a^2 \qquad K_+ = \frac{1}{2}a^{\dagger 2} \tag{23}$$

Where $K_\pm = \pm i(K_1 \pm iK_2)$ with the commutation relations

$$[K_3,K_\pm] = \pm K_\pm \qquad [K_-,K_+] = 2K_3 \tag{24}$$

Since the manifold generated by the action of $SU(1,1)$ stays the same, we don't have to recalculate the elements of the CGT. The only things which differs from the previous section are the expectation values

$$(\Delta K_1)_0^2 = \langle n| K_1^2 |n\rangle = \frac{1}{8}\left(n^2+n+1\right) \tag{25}$$

$$\langle K_3\rangle_0 = \frac{1}{2}(n+\frac{1}{2}) \tag{26}$$

Therefore the determinant of the CGT is

$$\det T_{jk}^{(n)} = \frac{1}{64}(n-1)n(n+1)(n+2)\sinh\theta \tag{27}$$

So it is equal to zero in the case of $n=0$, $n=1$ and $\theta=0$.

IV. QGT FOR THE GENERAL SU(1,1) CSS

First let's summarize all the relevant information about the $SU(1,1)$ CS-s[**?**], Perelomov. They are known to be in the UIR's known as the discrete positive series $\mathcal{D}^+(k)$, where k is the so-called Bargmann index. It appears in the eigenvalues of the Casimir-operator.

$$\begin{aligned} C_2 &= K_3^2 - K_1^2 - K_2^2 \\ &= K_3^2 - \frac{1}{2}\left(K_+K_- + K_-K_+\right) = \frac{1}{4}\left(\Delta^2-1\right) \end{aligned} \tag{28}$$

Where $\Delta = a^\dagger a - b^\dagger b$ is the occupation number difference operator between the two modes. In $\mathcal{D}^+(k)$ the Bargmann index is $k = 1/2, 1, 3/2, ...$, and the K_3, C_2 common basis elements are indexed by $|m,k\rangle$ where $(m = 0, 1, 2, ...)$. The operators act the following way on the eigenstates:

$$C_2 |m,k\rangle = k(k-1)|m,k\rangle \tag{29a}$$

$$K_3 |m,k\rangle = (m+k)|m,k\rangle \tag{29b}$$

$$K_+ |m,k\rangle = \sqrt{(m+1)(m+2k)}\,|m+1,k\rangle \tag{29c}$$

$$K_- |m,k\rangle = \sqrt{m(m+2k-1)}\,|m-1,k\rangle \tag{29d}$$

Therefore the general $SU(1,1)$ basis element can be constructed by acting on the ground state $|0,k\rangle$ with the K_+ operator

$$|m,k\rangle = \sqrt{\frac{\Gamma(2k)}{m!\Gamma(2k+m)}}(K_+)^m |0,k\rangle \tag{30}$$

It can be seen that in the Schwinger-type two-boson realization the indices are the following

$$k = \frac{1}{2}\left(|n_1-n_2|+1\right) \in \frac{1}{2}\mathbb{Z}^+ \tag{31a}$$

$$m = \frac{1}{2}\left(n_1+n_2-|n_1-n_2|\right) \in \mathbb{N} \tag{31b}$$

The $SU(1,1)$ CS-s according to Perelomov are

$$|\eta,k\rangle = O(\alpha)|0,k\rangle \tag{32a}$$

$$O(\alpha) = \exp\left[\alpha K_+ - \overline{\alpha}K_-\right], \qquad \alpha = \frac{\theta}{2}e^{-i\varphi} \tag{32b}$$

Where $\eta = -\tanh\frac{\theta}{2}e^{-i\phi}$

$$|\eta,k\rangle = (1-|\eta|)^k \sum_{m=0}^{\infty}\sqrt{\frac{\Gamma(m+2k)}{m!\Gamma(2k)}}\eta^m |m,k\rangle \tag{33a}$$

$$|\eta,q\rangle = (1-|\eta|)^{(1+q)/2} \sum_{n_2=0}^{\infty}\sqrt{\frac{(n_2+q)!}{n_2!q!}}\eta^{n_2} |n_2+q,n_2\rangle \tag{33b}$$

Where $q = |n_1-n_2|$ is the occupation number difference.

Again the only things which differs from the previous section concerning the CGT are the expectation values

$$(\Delta K_1)_0^2 = \langle m,k| K_1^2 |m,k\rangle = \frac{1}{2}(m^2+2mk+k) \tag{34}$$

$$\langle K_3\rangle_0 = m+k \tag{35}$$

Therefore the determinant of the CGT is

$$\det T_{jk}^{(n)} = \frac{1}{4}m(m+1)(2k+m-1)(2k+m)\sinh\theta \tag{36}$$

V. QGT OF AN $SU(2)$ QUBIT EPR STATE

Let's consider the state:

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle\otimes|1\rangle + |1\rangle\otimes|0\rangle\right) \tag{37}$$

Which is an entangled 2 oscillator state or in other words, an entangled 2 qubit EPR state embedded in 2 bosonic Fock spaces.

We want to calculate the QGT of this state assuming that our system has an $SU(2)$ symmetry. We will do this calculation in 2 ways. First we use equation (5). For that we need to calculate $U^\dagger\partial_\theta U$ and $U^\dagger\partial_\varphi U$. Since $SU(2)$ is a compact group, there exist a finite dimensional representation for it's algebra that is hermitian, for example $J_i = \frac{1}{2}\sigma_i$. Therefore we write U instead of O for the CS operator since it is now a unitary operator. We write calculated the following expressions in this algebra, but we'll only need the first 2 equations in the first proof.

$$U^\dagger \partial_\theta U = i \sin\varphi J_1 - i\cos\varphi J_2 \tag{38a}$$

$$U^\dagger \partial_\varphi U = 2i\sin^2\frac{\theta}{2} J_3 + i\sin\theta\left(\cos\varphi J_1 + \sin\varphi J_2\right) \tag{38b}$$

$$\partial_j U^\dagger U = -U^\dagger \partial_j U \quad (j=\theta,\varphi) \tag{38c}$$

$$\partial_\theta U^\dagger \partial_\theta U = \frac{1}{4} \tag{38d}$$

$$\partial_\theta U^\dagger \partial_\varphi U = \frac{i}{2} J_3 - i\sin^2\frac{\theta}{2}\left(\cos\varphi J_1 + \sin\varphi J_2\right) \tag{38e}$$

$$= -\partial_\varphi U^\dagger \partial_\theta U \tag{38f}$$

$$\partial_\varphi U^\dagger \partial_\varphi U = \sin^2\frac{\theta}{2} \tag{38g}$$

Now we specify that we will use the 2 oscillator Schwinger realisation fr the $SU(2)$ group aswell:

$$J_1 = \frac{1}{2}\left(a^\dagger b + a b^\dagger\right) \tag{39a}$$

$$J_2 = \frac{1}{2i}\left(a^\dagger b - a b^\dagger\right) \tag{39b}$$

$$J_3 = \frac{1}{2}\left(a^\dagger a - b^\dagger b\right) \tag{39c}$$

$$\tag{39d}$$

The action of these operators on the state $|\psi^+\rangle$ is the following.

$$J_1 |\psi^+\rangle = \frac{1}{2}|\psi^+\rangle \tag{40a}$$

$$J_2 |\psi^+\rangle = \frac{1}{2i}|\psi^-\rangle \tag{40b}$$

$$J_3 |\psi^+\rangle = \frac{1}{2}|\psi^-\rangle \tag{40c}$$

$$|\psi^\pm\rangle = \frac{1}{\sqrt{2}}\left(|10\rangle \pm |01\rangle\right) \tag{40d}$$

The sum in the definition of $T^{(+)}_{ij}$ runs over the orthocomplement of the $|\psi\rangle$ state's span. One possibility is

$$|m\rangle \in \left\{|00\rangle, |\psi^-\rangle, |22\rangle, |33\rangle, ...\right\} \tag{41}$$

These terms are all orthogonal to each other and are elements of the different irreps of $SU(2)$ since in the Schwinger realisation

$$|j,m\rangle = |n_a\rangle \otimes |n_b\rangle \implies j = \frac{1}{2}(n_a + n_b), \quad m = \frac{1}{2}(n_a - n_b) \tag{42}$$

From all this, it is straightforward to calculate the matrix elements:

$$T^{(+)}_{\theta\theta} = \frac{1}{4}\cos^2\varphi \tag{43a}$$

$$T^{(+)}_{\theta\varphi} = -\frac{1}{2}\left(\frac{1}{2}\sin\theta\sin\varphi + i\sin^2\frac{\theta}{2}\right)\cos\varphi \tag{43b}$$

$$T^{(+)}_{\varphi\theta} = -\frac{1}{2}\left(\frac{1}{2}\sin\theta\sin\varphi - i\sin^2\frac{\theta}{2}\right)\cos\varphi \tag{43c}$$

$$T^{(+)}_{\varphi\varphi} = \sin^2\frac{\theta}{2} - \frac{1}{4}\sin^2\theta\cos^2\varphi \tag{43d}$$

The second calculation comes from an equivalent definition of $T^(\psi)_i j$.

$$T^{(+)}_i j = \langle\psi|\,\partial_i U^\dagger \partial) j U\,|\psi\rangle - \langle\psi|\,\partial_i U^\dagger U\,|\psi\rangle\,\langle\psi|\,U^\dagger \partial_j U\,|\psi\rangle \tag{44}$$

By using the other expressions in equation (38a) we get back the same matrix elements.

Let's evaluate the determinant of this embedded qubit EPR state's QGT determinant, which tells us about it's uncertainty.

$$\begin{aligned}\det T^{(+)} &= \frac{1}{4}\cos^2\varphi\left(\sin^2\frac{\theta}{2} - \frac{1}{4}\sin^2\theta\cos^2\frac{\theta}{2}\right) \\ &\quad - \frac{1}{4}\cos^2\varphi\left(\frac{1}{4}\sin^2\theta\sin^2\varphi + \sin^2\frac{\theta}{2}\right) \\ &= \frac{1}{4}\cos^2\varphi\left(\sin^2\frac{\theta}{2} - \frac{1}{4}\sin^2\theta - \sin^4\frac{\theta}{2}\right) = 0\end{aligned} \tag{45}$$